

Development of an LLM-supported workflow for semantic metadata extraction from catalytic experiments

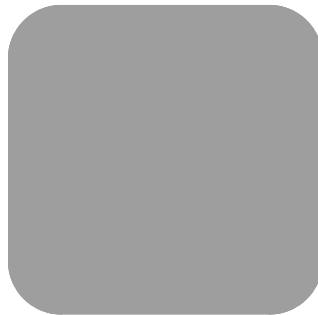
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Second examiner: Prof. Dr.-Ing. S. Lucia

Supervisor: M.Sc. Marc Völkenrath

Supervisor: M.Sc. Hendrik Borgelt

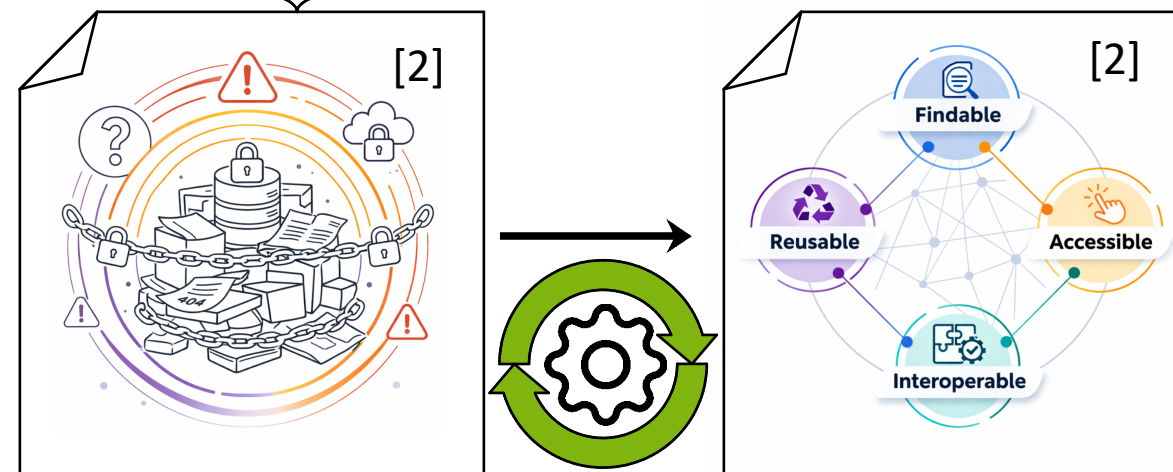
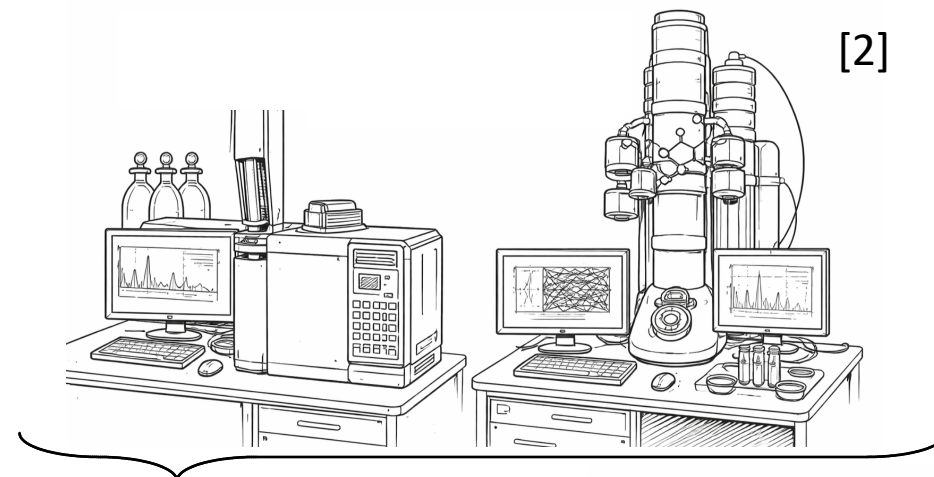
```
"Company": "Bruker Nano GmbH Berlin, Germany",  
"Software": "Esprit",  
"Spectrum": 1,  
"Date": "05.05.2022",  
"Primary energy": {  
  "value": 200.0,  
  "unit": "http://qudt.org/vocab/unit#kJoule"  
},  
"Tilt angle": {  
  "value": 0.0,  
  "unit": "http://qudt.org/vocab/unit#DegreeAngle"  
},  
"Take off angle": {  
  "value": 22.0,  
  "unit": "http://qudt.org/vocab/unit#DegreeAngle"  
},  
"Azimut angle": {  
  "value": 45.0,  
  "unit": "http://qudt.org/vocab/unit#DegreeAngle"  
},
```



Motivation^[1]

- **Sources** for data in catalysis research
 - Experimental data
 - Computational data sources
- Data **without metadata**
 - Hard to reproduce
 - Not usable in data driven discoveries
- **Challenges** in metadata capture ^[1]
 - Manual work → low adoption
 - Prone to inconsistency

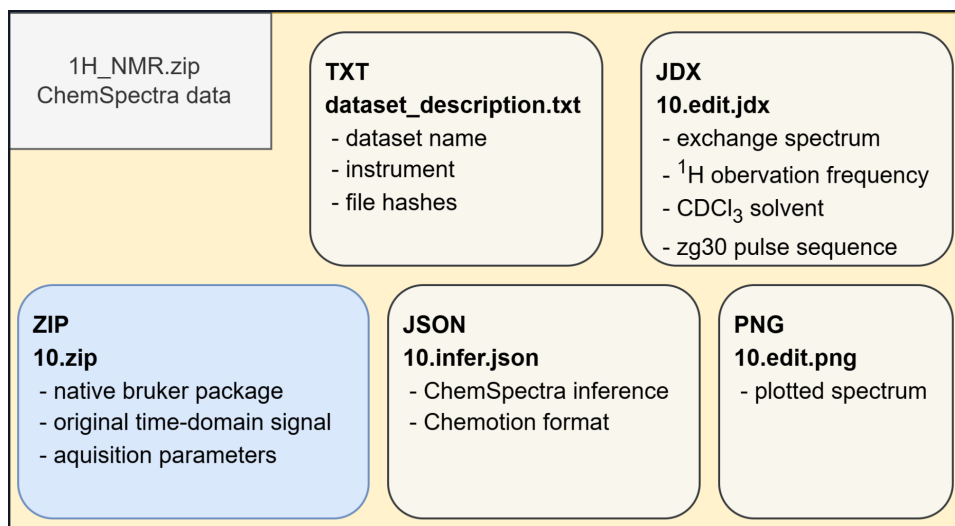
→ Automation drives metadata consistency and adoption



[1] P. S. F. Mendes, N. López, S. De, R. R. Fushimi, R. Yildirim, and A. Trunschke, "Data as a Key Resource in Catalysis: A Community Account," ChemCatChem, 17(24), e01226 (2025). DOI: 10.1002/cctc.202501226.

[2] Generated with ChatGPT

Data Sources and Formats



“

```
##TITLE= Parameter file
##JCAMPDX= 5.0
$ 2021-09-24 13:20:39.908 +0200
$ PROBHD= 5 mm PABBO BB/19F-1H/D
...
##$ NUC1=          1H
...
##$ SF01=          500.13
...
##$ SOLVENT=       CDCl3
```

has_aquisition_nucleus [2]

rdfs:label: aquisition nucleus
rdfs:type: NMRCV:1400083
value: 1H

has_spectrometer_frequency

rdfs:label: irradiation frequency
rdfs:type: NMRCV:1400026
qudt:unit: MegaHZ
value: 500.13

has_solvent

rdfs:label: NMR solvent
rdfs:type: NMRCV:1000330
value: CDCl3

- Multiple file types
- Data scattered across dataset
- Relevant lines between irrelevant lines
- Local terminology
- Machine actionable data
- Common terminology

→ Most datasources are information-rich, but not self-describing

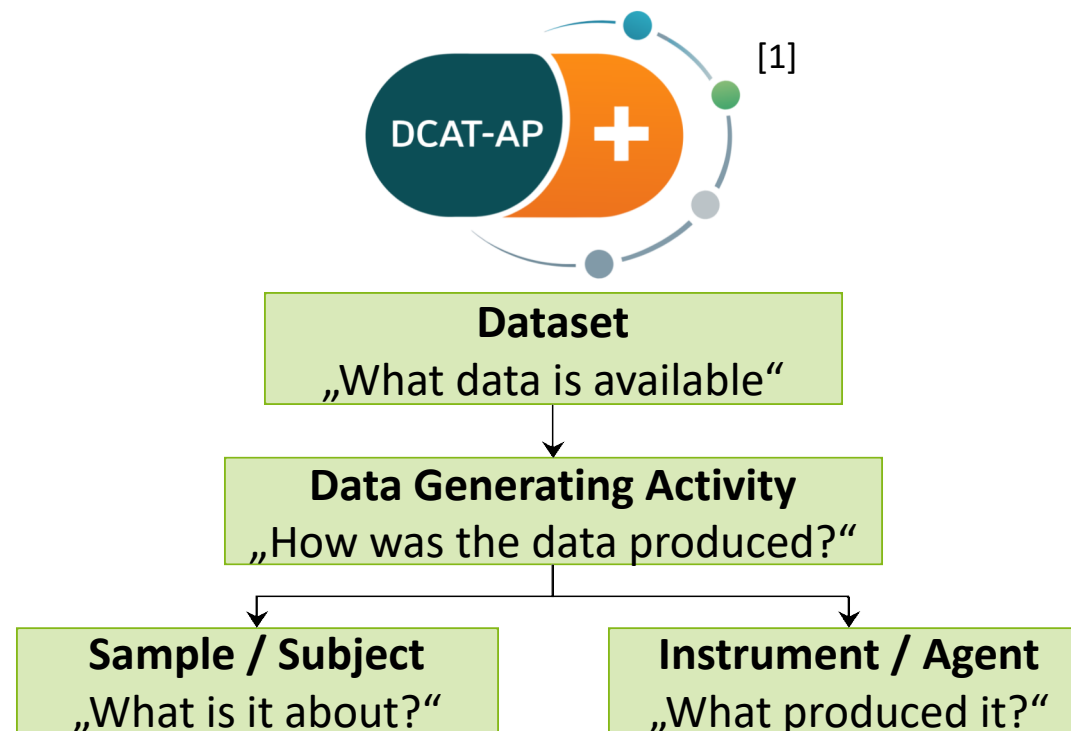
[1] Chemotion Repository, <https://doi.org/10.14272/QGYHKEGFODSECR-UHFFFAOYSA-N/CHMO0000591>

[2] nmrCV v1.1.0; Schober et al., Anal. Chem. 2018, 90, 649–656. doi:10.1021/acs.analchem.7b02795

DCAT-AP+ as Semantic Data Model

„DCAT-AP Plus Links to Use-case Specific context „

- Provenance Ontology (PROV-O) alignment
 - Domain agnostic data modeling for heterogenous catalysis data
 - Foundation for domain-specific models
- Generic attributes
 - Provide structure for descriptive information
 - Parent slots for domain specific attributes
- Flexible classification
 - Entity typing with OWL classes or SKOS concepts
 - Target slots for vocabulary annotation



[1] Philip Strömert et al. nfdi-de/dcat-ap-plus: Release 0.1.0rc4. Version v0.1.0rc4. Mar. 2026.
DOI: 10.5281/zenodo.18921353.

Large Language Models (LLMs)

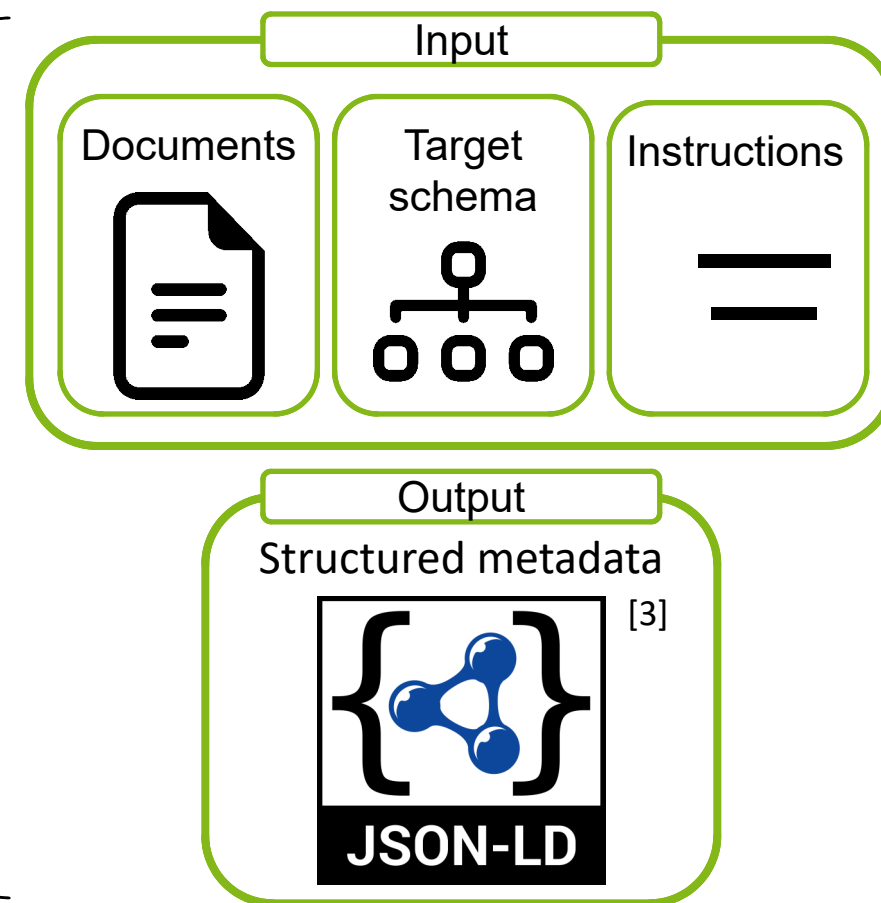
- Why an LLM?
 - Processes heterogeneous textual content
 - Recognizes metadata with varying terminology^[1]
 - Structured output
- Local models with Ollama
 - Fit on consumer hardware (<8GB VRAM)
 - Independence from external providers
- Challenges
 - No specialized domain capabilities
 - <8k input token per processing step

→ Extraction needs to be executed in chunks

[1] B. W. Walls and S. Linic, ACS Catal. 2025, 15, 14751–14763. DOI: 10.1021/acscatal.5c03844.



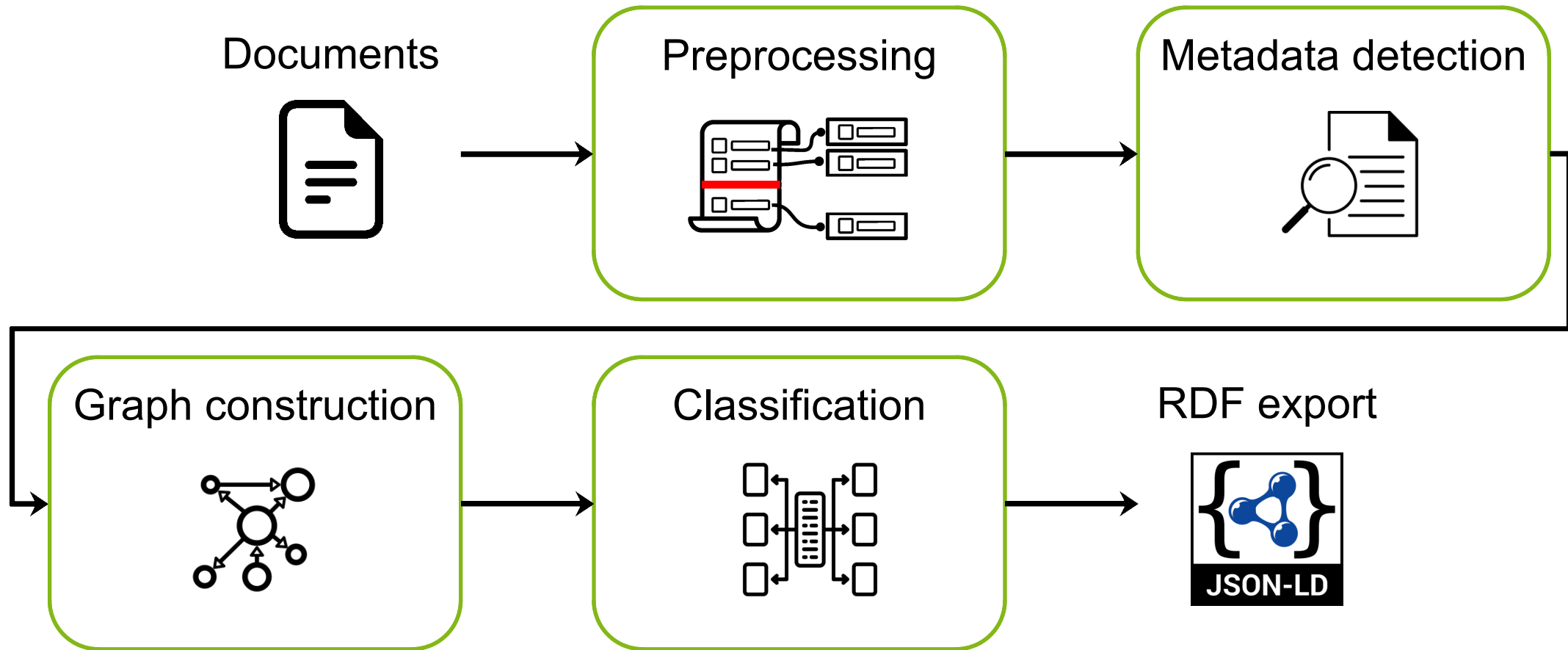
[2]



[2] Ollama.com

[3] JSON-LD Community Group, json-ld.org

Workflow

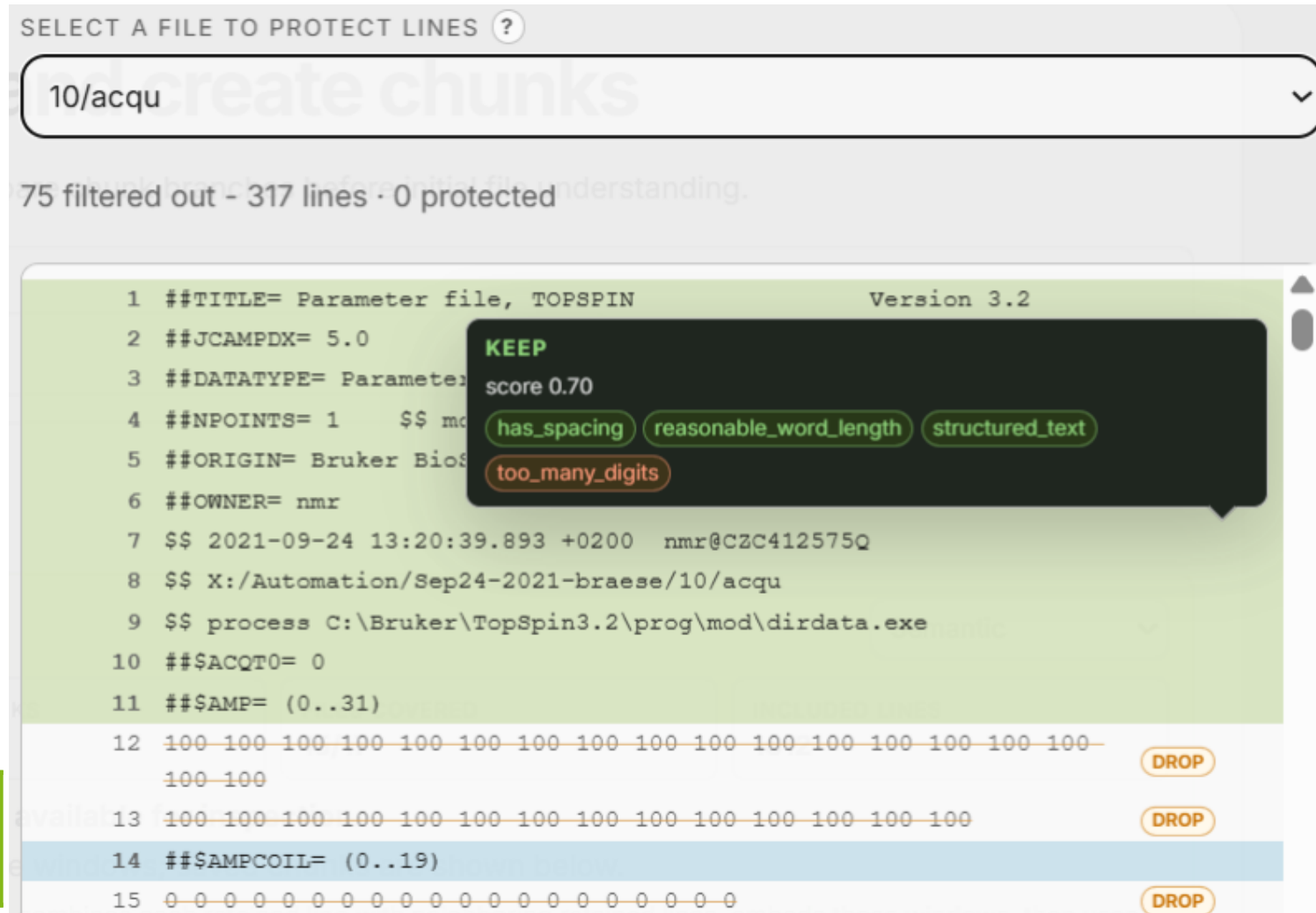


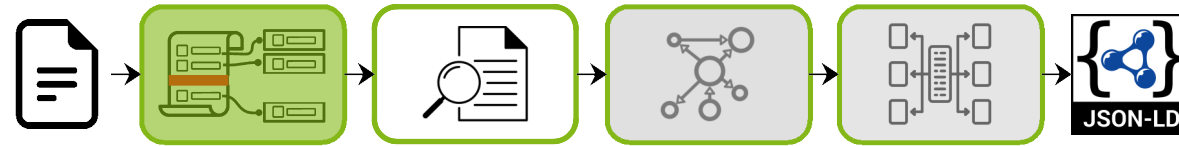
Preprocessing

Process

-
- ```
graph TD; 1((1)) --> 2((2)); 2 --> 3((3)); 3 --> 4((4));
```
- 1 Dataset with multiple files
  - 2 Collect text-based files
  - 3 Text quality filter
  - 4 Split large files in chunks

→ Prepared chunks for metadata detection





## Metadata Detection

- Per chunk processing
  - Chunks sizes determine granularity
  - Chunks are connected to dataset and file context
- Dataset context
  - File names and small summaries
  - General overview of dataset content
- Evidence note extraction
  - First LLM produces list of evidence notes
  - Second LLM classifies evidence

### „Evidence“

Source-linked dataset content to support metadata claim

**Evidence note**

- category
- role
- claim
- evidence\_text



**Critic**

### Portable evidence

- grounded
- self-contained

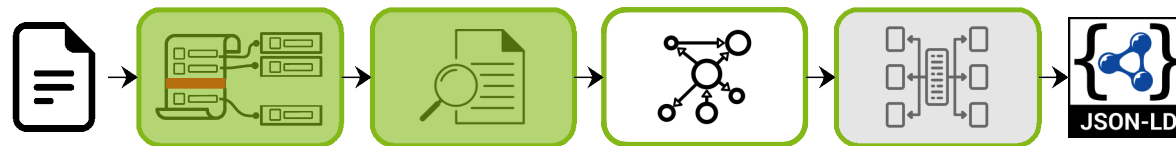
### Contextual evidence

- grounded
- dependent

### Rejected evidence

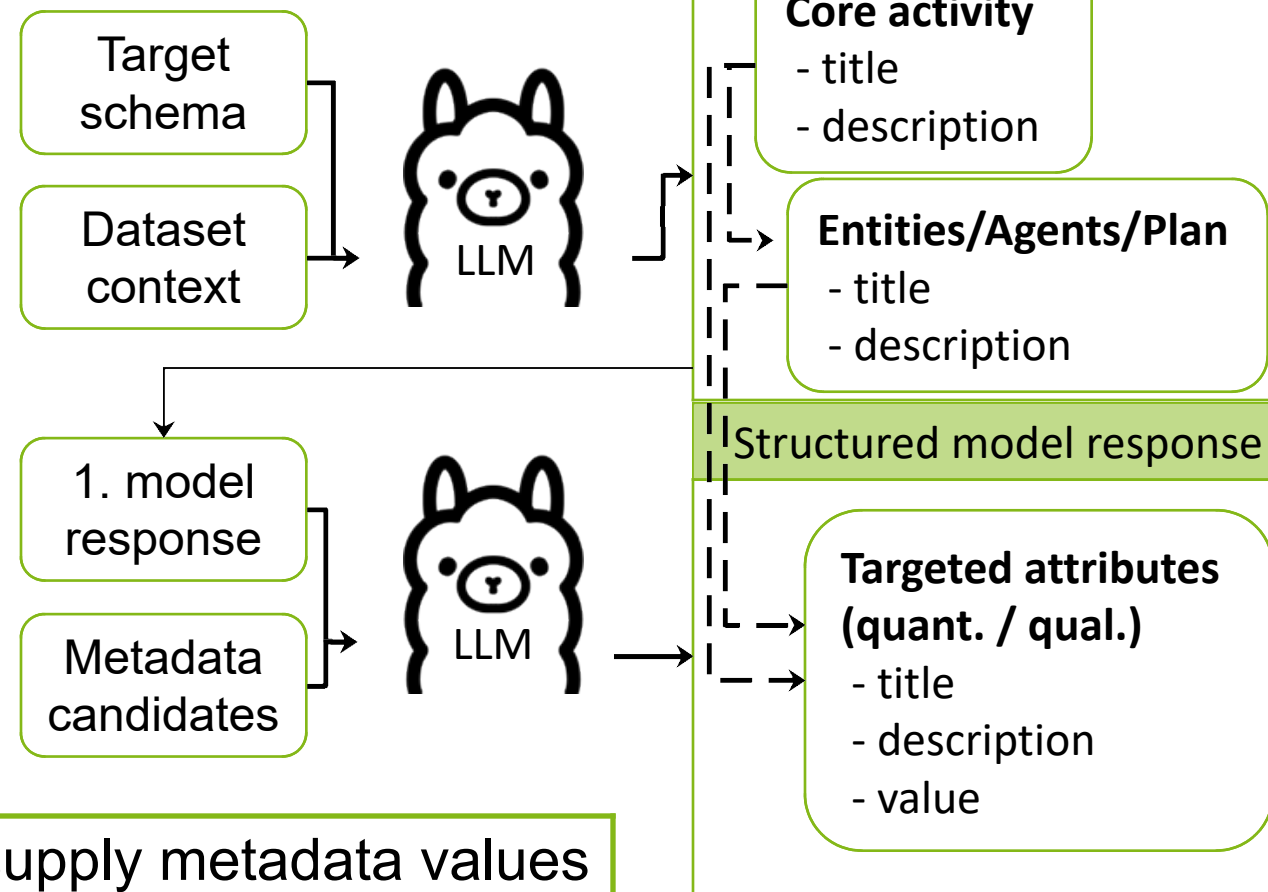
- ungrounded
- ambiguous

→ Evidence notes form retrievable set of metadata candidates



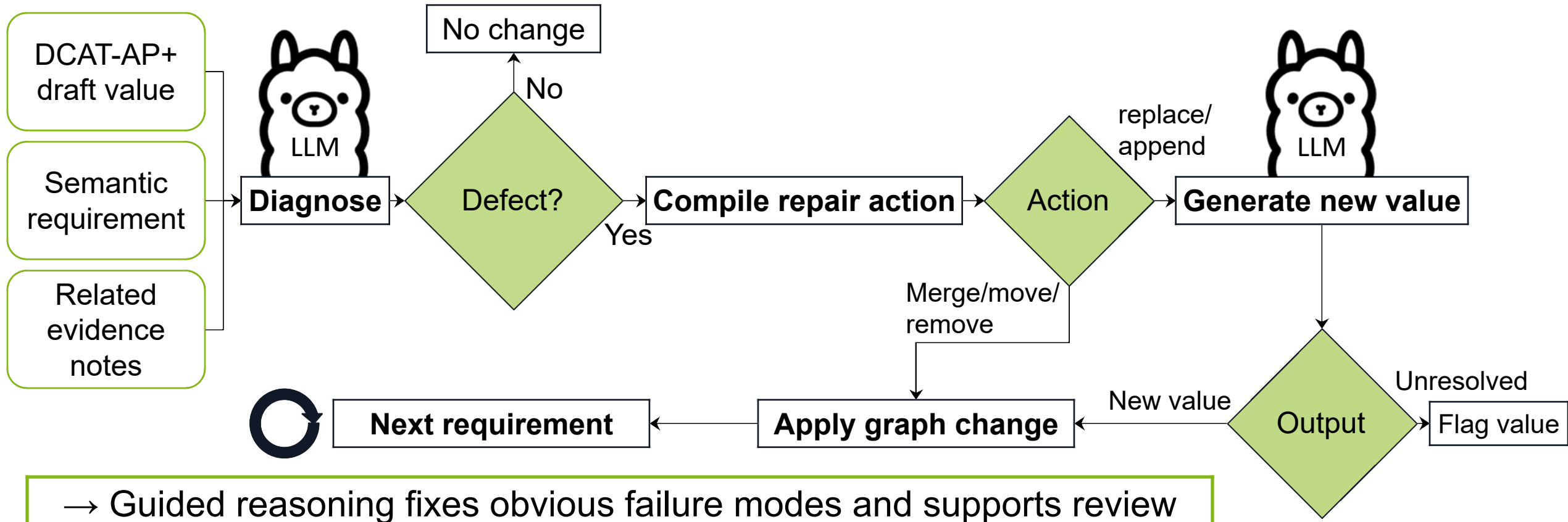
## Graph Construction - Provenance and Attributes

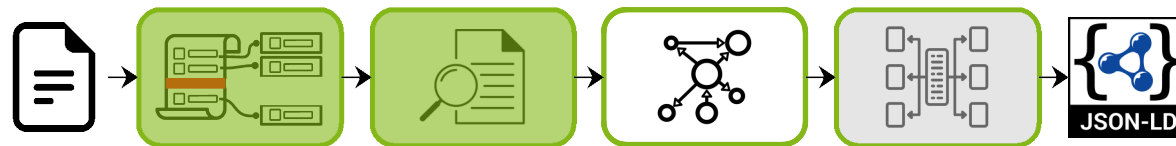
- Provenance core
  - Detect central activity
  - Add related entities, agents, plan, etc.
  - Build graph from DCAT-AP+ schema
  
- Attributes
  - Retrieve parent-specific metadata
  - Attach attributes to related parents



→ Provenance defines structure; attributes supply metadata values

## Semantic Validation and Reconstruction





## Semantic Validation and Reconstruction - Examples

„Aquisition settings belong to the data-generating activity.“

**Before**

```
"evaluated_entity": {
 "title": "measured sample",
 "number of scans": 16
}
```

**Defect:** wrong\_attribute\_parent

move

**After**

```
"was_generated_by": {
 "title": "1D 1H NMR acquisition",
 "number of scans": 16
}
```

Value retained, only parent assignment changes

„A numeric range is represented as separate minimum and maximum attributes.“

**Before**

```
"value": "373.96-3997.45",
"type": "Wavenumber range",
"unit": "1/CM"
```

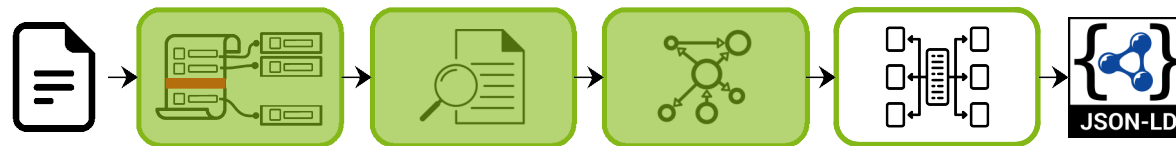
**Defect:** bad\_range

append minimum + maximum;  
remove collapsed range

**After**

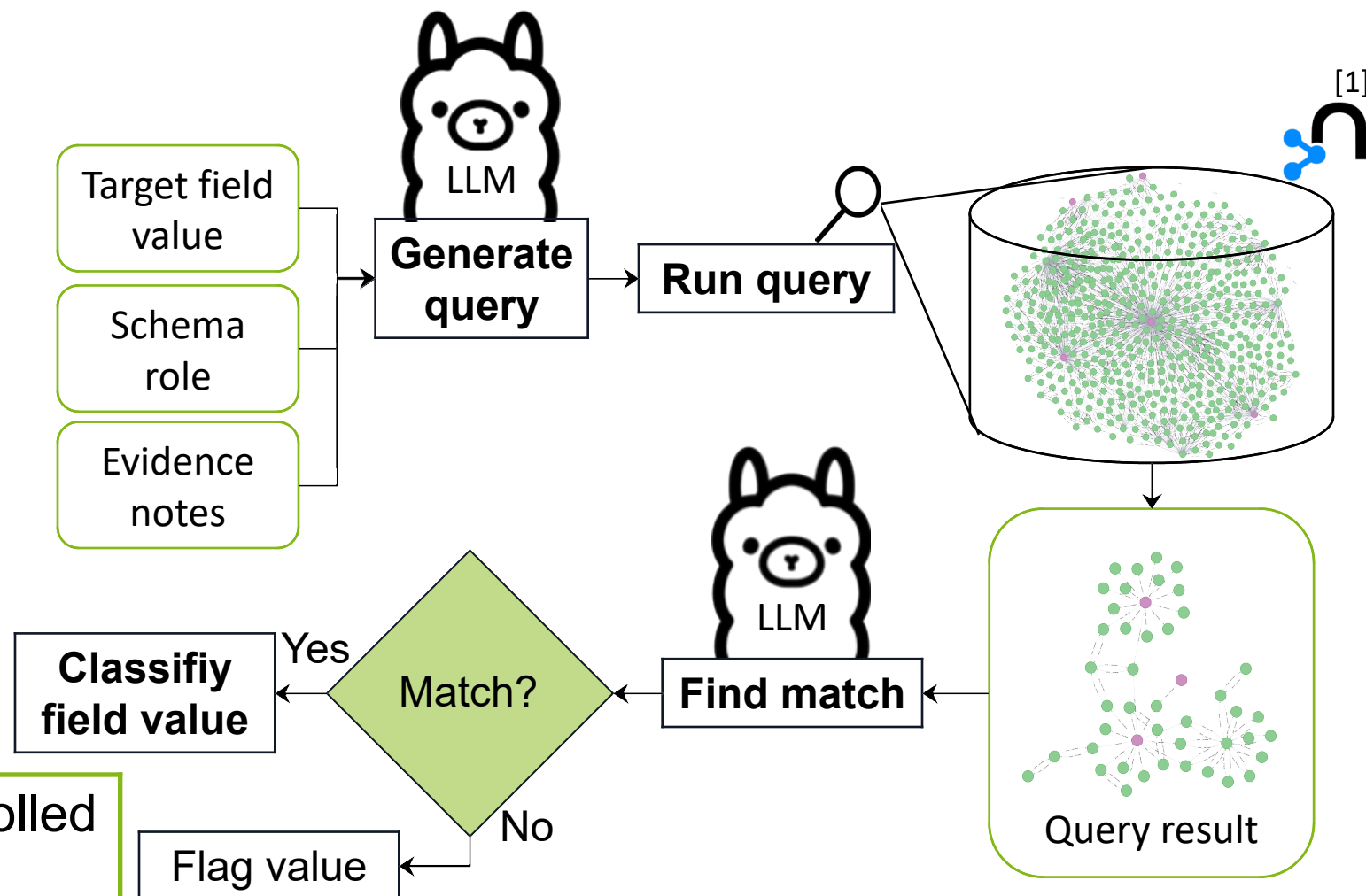
```
{ "value": 373.96, "type": "minimum wavenumber", "unit": "1/CM" },
{ "value": 3997.45, "type": "maximum wavenumber", "unit": "1/CM" }
```

Multiple bounded actions reconstruct the range



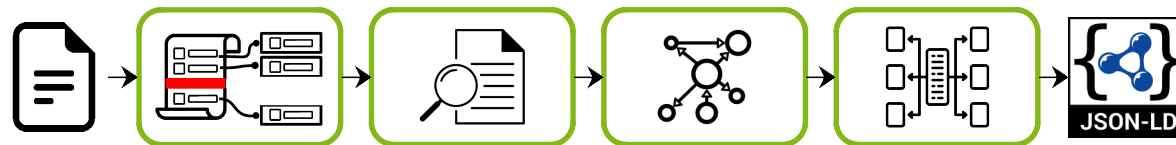
## Vocabulary Grounding

- Hybrid query strategy
  - Vector similarity score
  - Matches in labels
  - Top k nodes → graph traversal
- Matching
  - Compare value with retrieved candidates
  - Assign best match
  - Flag if no suitable match



→ Structured integration of controlled vocabulary terms

[1] neo4j.com



## Graphical user interface

### Flagged requirement with its evidence

**Dataset (8)**

- id: 2e38a6a6
- title [1]
- description [1]
- was\_generated\_by [1]
- DataGeneratingActivity 1 (10)
  - title [1]
  - description [1]
  - realized\_plan: Plan (2)
  - carried\_out\_by [1]
  - AgenticEntity 1 (5)
    - has\_qualitative\_attrib
      - QualitativeAttribute
        - type: DefinedTe
      - type: DefinedTerm (3)
    - evaluated\_entity [1]
    - EvaluatedEntity 1 (4)

**DATASET**

**Patch review**

**Missing** was\_generated\_by.0.has\_quantitative\_attribute.0.type  
missing, non\_enriched

**ISSUES**

- missing
- non\_enriched

**Non-enriched** was\_generated\_by.0.has\_quantitative\_attribute.0.type  
missing, non\_enriched

**ISSUES**

- missing
- non\_enriched

**ACQT0 mislabelled as a duration**

**Missing** was\_generated\_by.0.type  
missing, non\_enriched

**ISSUES**

open field

### Navigate to the exact field and correct it

**Curated document** RDF explorer

**TITLE**  
Acquisition Time (ACQT0) **Edit: acquisition-time offset**

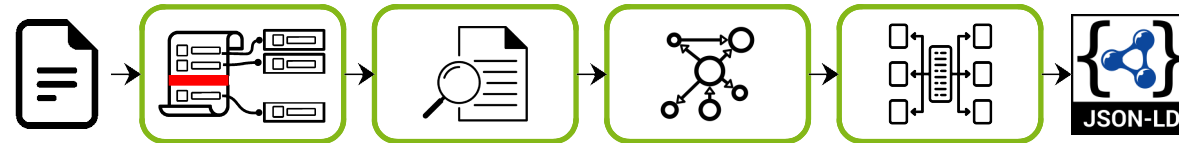
**DESCRIPTION**  
The acquisition time ACQT0 recorded during the data acquisition activity.

**VALUE**  
-2,52100840336134

**HAS\_QUANTITY\_TYPE**  
http://qudt.org/vocab/quantitykind/Time

**UNIT**  
http://qudt.org/vocab/unit/SEC

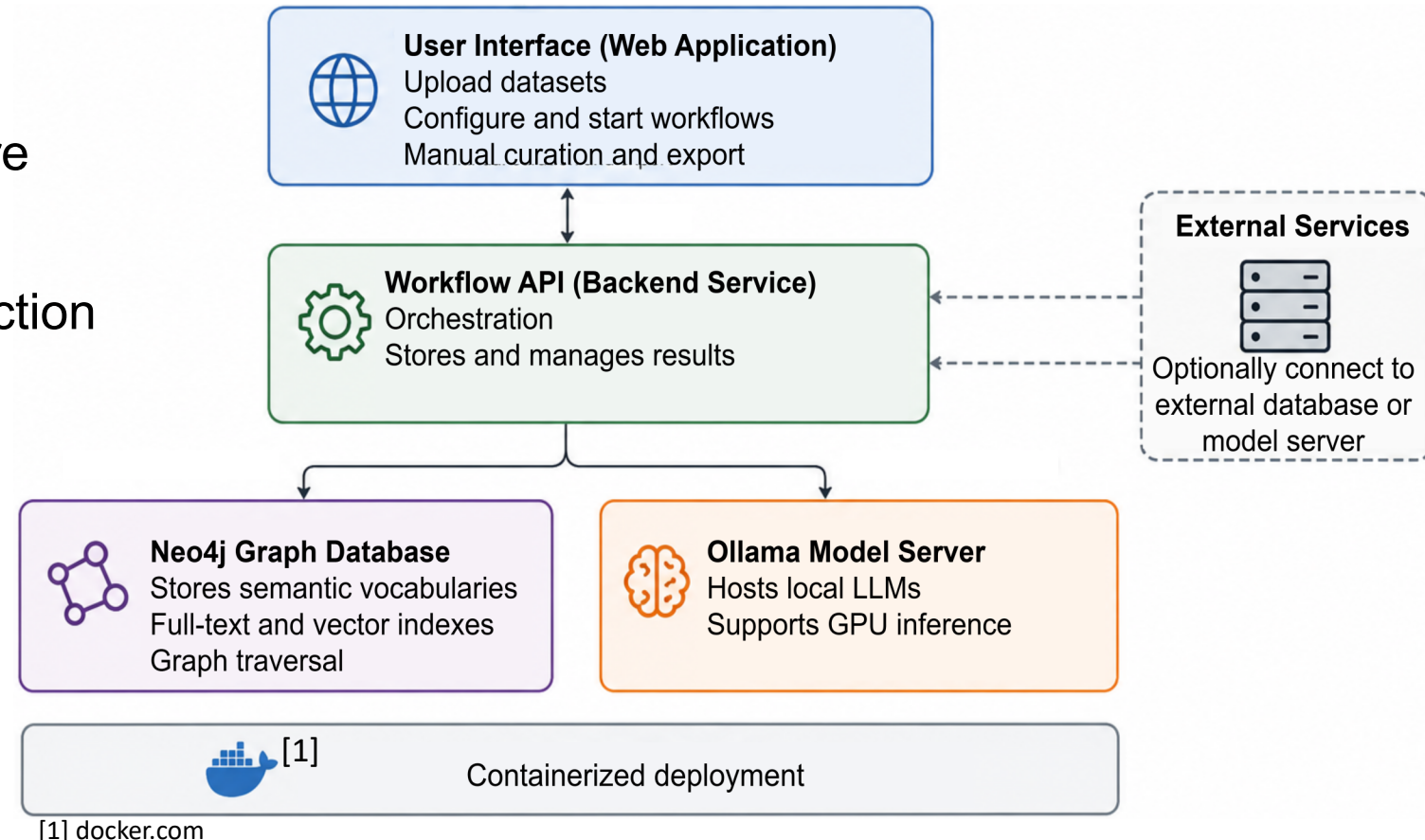
Live-Demo possible



## Result

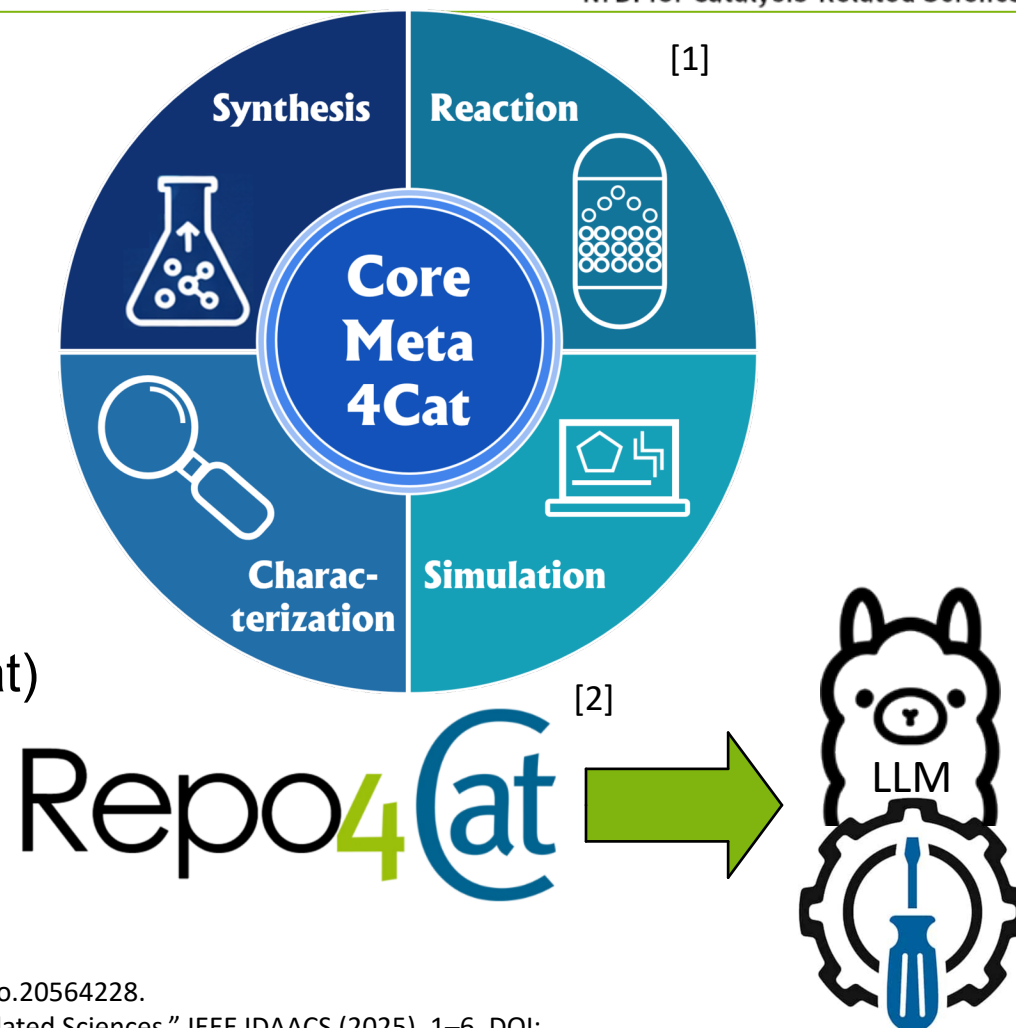
- End-to-end prototype
  - Modular workflow components
  - Runs locally on consumer hardware
- Assisted RDF generation
  - Requirement driven graph construction
  - Supports transparent review
- Application-profile development
  - Generates DCAT-AP+ source data
  - Reveals modeling gaps

→ Closes methodological gap for semantic metadata extraction



## Outlook

- Metadata extraction
  - Domain agnostic baseline misses claims
  - Unstable retrieval in downstream steps
    - Domain aware evidence labeling and categorization
  
- RDF graph construction
  - Domain-specific semantic requirements (CoreMeta4Cat)
  - Create source data for model fine tuning (Repo4Cat)
    - Specialized models for semantic graph creation



[1] Borgelt, H.; Neyyathala, A.; Khatamirad, M. CoreMeta4Cat, v0.0.1rc2, Zenodo (2026). DOI: 10.5281/zenodo.20564228.

[2] Kushnarenko, V. et al. "Repo4Cat: How to Setup, Configure and Operate a Data Repository for Catalysis-Related Sciences," IEEE IDAACS (2025), 1–6. DOI: 10.1109/IDAACS68557.2025.11322400.



- Simple Fließbild als Motivation (input → output)
- → FAIR erklären ohne FAIR zu erklären
- Daten, Vokabulare (GC-MS), Datensätze
- ÜBERLEITUNG (Blackbox, LLMs, etc..)
- Grober Workflow als Fließbild
- → Bausteine vorstellen + Fazit und vergleichen
- → (Zeit für Entwicklung pro Baustein)
- → Saubere Bewertung fürs Publikum ist relevant
- → semantische Requirements als Versuch zur Quantifizierung
- **Positives:** privacy konform, UI und nachvollziehbarkeit, semantic requirements als KPI um Benchmarking voranzutreiben, user
- Moduabhängige Problematiken zusammenfassen und übergeordnete Herausforderungen anzeigen (keine annotierten Daten, unzulängliche Ressourcen etc..)

## Example Vector Index Query

**POST** /api/v1/index/query Query Index

Parameters

| Name                                              | Description                                                           |
|---------------------------------------------------|-----------------------------------------------------------------------|
| <b>index_name</b> * required<br>string<br>(query) | skos__Concept_vector_index                                            |
| <b>query</b> * required<br>string<br>(query)      | gas chromatography mass spectrometry                                  |
| <b>top_k</b><br>integer<br>(query)                | The number of top results to return<br>5<br><small>minimum: 1</small> |

Execute

### Response body

```
{
 "entities": [
 {
 "dcterms_identifier": "voc4cat_0000130",
 "skos_prefLabel": "gas chromatography",
 "skos_historyNote": "0000-0002-6242-2167 Nikolaos G. Moustakas created the resource",
 "skos_altLabel": "GC",
 "skos_definition": "An analytical technique that separates and analyzes the components of a gas mixture by passing it through a stationary phase.",
 "uri": "https://w3id.org/nfdi4cat/voc4cat_0000130",
 "labels": [
 "Resource",
 "skos_Concept"
]
 },
 {
 "dcterms_identifier": "voc4cat_0000137",
 "skos_prefLabel": "mass spectrometry",
 "skos_historyNote": "0000-0002-6242-2167 Nikolaos G. Moustakas created the resource",
 "skos_altLabel": "MS",
 "skos_definition": "An analytical technique which determines the mass-to-charge ratio of ions in a sample to give information about the sample.",
 "uri": "https://w3id.org/nfdi4cat/voc4cat_0000137",
 "labels": [
 "Resource",
 "skos_Concept"
]
 }
]
}
```

## Example Vector Index Query 2

**POST** /api/v1/index/query Query Index

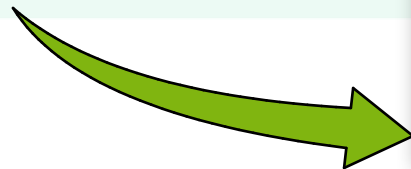
Parameters

| Name                                              | Description                                                            |
|---------------------------------------------------|------------------------------------------------------------------------|
| <b>index_name</b> * required<br>string<br>(query) | qudt__Unit_vector_index                                                |
| <b>query</b> * required<br>string<br>(query)      | 40 °C/min                                                              |
| <b>top_k</b><br>integer<br>(query)                | The number of top results to return<br>20<br><small>minimum: 1</small> |

Execute

### Response body

```
{
 "entities": [
 {
 "qudt__uneceCommonCode": "H13",
 "qudt__iec61360Code": "0112/2///62720#UAA037",
 "qudt__symbol": "°C/min",
 "qudt__ucumCode": "Cel.min-1",
 "qudt__conversionMultiplier": 0.016666666666666666,
 "rdfs__label": "Degree Celsius per Minute",
 "dcterms__description": "$\\textit{Degree Celsius per Minute}$ is a unit for 'Temperature Per Time' expressed as $degC / m$.",
 "qudt__informativeReference": "https://cdd.iec.ch/cdd/iec62720/iec62720.nsf/Units/0112-2---62720%23UAA037",
 "qudt__conversionMultipliersN": 0.016666666666666666,
 "uri": "http://qudt.org/vocab/unit/DEG_C-PER-MIN",
 "qudt__expression": "$degC / m$",
 "labels": [
 "Resource",
 "qudt__Unit"
]
 },
 {
 "qudt__uneceCommonCode": "J24",
 "qudt__iec61360Code": "0112/2///62720#UAA045",
 "qudt__symbol": "°F/min",
 "dcterms__description": "$\\textit{Degree Fahrenheit per Minute}$ is a unit for 'Temperature Per Time' expressed as $degF / m$.",
 "rdfs__label": "Degree Fahrenheit per Minute",
 "qudt__conversionMultiplier": 0.009259259259259259,
 "qudt__ucumCode": "[degF].min-1"
 }
]
}
```



## F1 Score

$$\text{Precision} = \frac{TP}{(TP + FP)}$$

$$\text{Recall} = \frac{TP}{(TP + FN)}$$

$$\text{F1 Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{(\text{Precision} + \text{Recall})}$$